

Axiomatic Faith: What Rudin Taught me About God

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When I was applying to college, I wrote that I wanted to study mathematics because it was the “closest thing we had to truth”. History could be revised, literature interpreted, politics endlessly contested, but mathematics seemed to exist outside of us, a clean structure we discovered rather than invented. One plus one equals two whether you are in Taipei, where I grew up, or New York, where I now study. The Pythagorean theorem held for the Greeks, and it holds for me. At the time that certainty was, frankly, the appeal. I wanted a discipline where I had the ability to be right.

A year into my degree at Columbia, I am less sure mathematics offers what I thought it did, and I think that is a good thing.

The shift came from an unlikely source. Alongside my problem sets in real analysis and abstract algebra, I started reading the Bible. Not for religious reasons, exactly, but because I was curious about a body of knowledge I had spent most of my education dismissing as the unfalsifiable cousin of the work I cared about. (And because LitHum required it) What I noticed surprised me. Theology, like mathematics, begins with premises that cannot be proved from within the system. You accept something, the existence of God, the authority of scripture, and reason outward. The conclusions are only as firm as the foundations.

Mathematics works exactly the same way. Every theorem I have ever proved rests on axioms: statements we declare to be true without proof. Euclidean geometry holds because we agree to Euclid's postulates; reject the parallel postulate and you get an entirely different geometry, equally consistent, describing the surface of a sphere. The integers exist because we accept Peano's axioms. Set theory's foundations were the subject of a hundred-year crisis that was never fully resolved — we chose a system and moved on. Mathematics is not truth handed down from the universe. It is a structure we built, brick by brick, on ground we agreed to call ground.

This should bother us more than it does. Why doesn't it?

Two reasons, I think. First, mathematics is not part of daily life for most people, so its foundations are never stress-tested the way a moral or political philosophy is. We argue about justice and meaning at the dinner table; we do not argue about the axiom of choice. Second, and harder, mathematics works. It models the physical world with such uncanny precision that the assumption of its objectivity feels less like an assumption than an observation. Bridges hold. Rockets land. Encryption protects our messages. When a system is that useful, its premises stop looking like premises and start looking like facts.

But they are still premises. And recognizing that has changed how I think about people whose foundational commitments differ from my own.

I used to find religious belief baffling. How could anyone build a life on something they could not prove? Now I notice I have done the same thing. I have committed to a system whose ground floor is, ultimately, an act of faith, a faith vindicated by results, but faith nonetheless. The mathematician and the theologian

are not as different as I once believed. We have both chosen our axioms; we are both trying to be honest about what follows.

This is not an argument against mathematics or science, or against the immense practical authority both have earned. It is an argument for honesty about what kind of authority that is. STEM students like me are quick to dismiss philosophy and theology as soft because their disagreements are visible, believers argue about doctrine, ethicists argue about trolley problems, and political philosophers argue about everything. Mathematicians do not appear to argue at all. But that is not because we have escaped the problem of foundations. It is because we agreed to stop having that argument a long time ago, somewhere down at the level of our axioms, and built upward from there in shared silence.

The contradictions in moral and political philosophy are loud because they are human and unavoidable; we live inside them. The contradictions at the base of mathematics are quiet because we have walled them off and stopped looking. Quiet is not the same as resolved.

What I have come to believe, after a year of proofs and a year of scripture, is that certainty in any discipline begins with a leap. The difference between fields is not whether the leap was taken, but whether we are willing to admit it.